Homework 7: Preliminaries to Scattering Theory

HW 7.1: In this question we will look into the state $i \to i$ transitions in time-dependent perturbation theory. It contains a lot of familiar results that nicely wraps up the perturbation theory chapter.

Working in the interaction picture,

$$|\psi_I(t)\rangle = e^{iH_0 t}\,|\psi(t)\rangle = \sum_n c_n(t)\;|n\rangle\,, \qquad c_i(0) = 1, \qquad c_{n\neq i}(0) = 0, \tag{1}$$

the Dyson expansion can be written as

$$c_n(t) = \delta_{ni} + c_n^{(1)}(t) + c_n^{(2)}(t) + \cdots \tag{2}$$

with

$$c_n^{(1)}(t) = -i\int_0^t dt_1\, e^{i(E_n-E_i)t_1} V_{ni}, \tag{3}$$

$$c_n^{(2)}(t) = -\sum_m \int_0^t dt_1 \int_0^{t_1} dt_2\, e^{i(E_n-E_m)t_1} e^{i(E_m-E_i)t_2} V_{nm}V_{mi}. \tag{4}$$

a) Show that to first order, the correction to the initial-state amplitude is just a phase shift and that the energy of the state is corrected just as in time-independent perturbation theory.

b) Starting from the Dyson expression with $n = i$, show that

$$c_i^{(2)}(t) = -\sum_m |V_{mi}|^2 \int_0^t dt_1 \int_0^{t_1} dt_2\, e^{i(E_i-E_m)(t_1-t_2)}. \tag{5}$$

What does the term with $m = i$ correspond to (see part (a))? For the remaining $m \neq i$ terms, perform the time integrals and identify the term that grows linearly with t. Show that the coefficient of this "secular" term reproduces the second-order energy shift of time-independent perturbation theory.

c) We now want to consider the case where the states $|m\rangle$ include a continuum of energies. In general, the energy denominator in part (b) must be regulated. One physical way to do this is to adiabatically switch on the perturbation from the infinite past, replacing $V \to Ve^{\eta t}$ with $\eta \to 0^+$ (this is the approach in Sakurai's book). Redo the calculation in part (b) with this replacement. You should find that the net effect on the energy denominator is just

$$\frac{1}{E_i - E_m} \;\longrightarrow\; \frac{1}{E_i - E_m + i\epsilon}, \qquad \epsilon = \eta \to 0^+. \tag{6}$$

Hint: Each V_I insertion now carries an extra factor of $e^{\eta t_k}$. The secular (linear-in-t) piece of $c_i^{(2)}(t)$ should look like

$$c_i^{(2)}(t) \supset -i\,t \sum_{m\neq i} \frac{|V_{mi}|^2}{E_i - E_m + i\epsilon}. \tag{7}$$

d) Now replace the sum over intermediate states by an integral weighted by a density of states $\rho(E)$. Use the identity,

$$\frac{1}{x + i\epsilon} = \mathcal{P}\left(\frac{1}{x}\right) - i\pi\,\delta(x), \tag{8}$$

where $\mathcal{P}$ denotes the Cauchy principal value, to separate the energy shift into a real and an imaginary part. Show that, for large t,

$$c_i(t) \simeq 1 - i\,\Delta_i\,t, \qquad \Delta_i = \Delta E_i - \frac{i}{2}\,\Gamma_i, \tag{9}$$

with

$$\Delta E_i = V_{ii} + \mathcal{P}\int dE\,\frac{|V_{iE}|^2\,\rho(E)}{E_i - E}, \tag{10}$$

$$\Gamma_i = 2\pi \int dE\,|V_{iE}|^2\,\rho(E)\,\delta(E_i - E) = 2\pi\,|V_{iE_i}|^2\,\rho(E_i). \tag{11}$$

What are the physical interpretations of ΔE_i and Γ_i? In particular, discuss what happens to the state $|i\rangle$ if $\Gamma_i \neq 0$.

> The results above show that an unstable state acquires a complex energy $\mathcal{E}_i = E_i + \Delta E_i - i\Gamma_i/2$. This has direct consequences for scattering and transition processes that proceed through $|i\rangle$ as an intermediate state.